

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA1407	Course Title:	COMPILER DESIGN
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	40% of CIA		
CO1 Statement:	Apply lexical analysis for token specification and recognition using LEX tool. (BL3)		
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Section / Batch:	Information Technology	Date:	26.07.2026.

Code of Conduct

I Arjun B (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: [Signature]

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Q. No	Problem Understanding (20)	Method Selection (20)	Solution Process (25)	Accuracy of Calculation (20)	Interpretation of Results (15)	Total Score (out of 100)
1	4 / 4	3 / 4	4 / 4	4 / 4	3 / 4	91.5
2	4 / 4	3 / 4	3 / 4	4 / 4	4 / 4	88.75
3	3 / 4	4 / 4	3 / 4	4 / 4	3 / 4	85
4	4 / 4	4 / 4	3 / 4	3 / 4	3 / 4	85
5	3 / 4	3 / 4	4 / 4	4 / 4	4 / 4	90
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	88
	/ 20	/ 20	/ 25	/ 20	/ 15	/ 100

1)

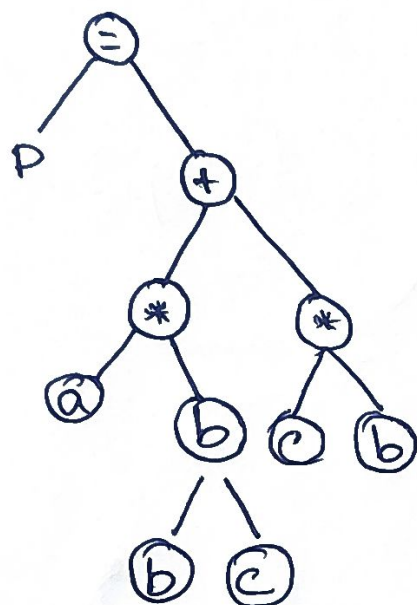
$$P = (a * b) + (b * c) * 2$$

Tokens:

$\langle P \rangle \Rightarrow \langle (\rangle \langle a \rangle \langle * \rangle \langle b \rangle \langle + \rangle \langle c \rangle \langle b \rangle \langle * \rangle \langle (\rangle \langle \rangle \rangle * 2$

$\langle id_1 \rangle \Rightarrow \langle (\rangle \langle id_2 \rangle \langle * \rangle \langle id_3 \rangle \langle + \rangle \langle (\rangle \langle id_4 \rangle \langle * \rangle \langle id_5 \rangle \langle \rangle \rangle * 2$

(ii) Syntax tree:



(iii) Intermediate code:

$$T_1 = id_4 * id_5$$

$$T_2 = T_1 * 2$$

$$T_3 = id_3 * id_4$$

$$T_4 = T_2 + T_3$$

$$T_A = id_1$$

(iv) Code Optimization:

$$t_1 = a * b$$

$$t_2 = b * c$$

$$P = t_1 + t_2 * 2$$

3) (i) Find the number of tokens in C program.

S.No	TOKENS
1	main
2	(
3	)
4	{
5	int
6	a
7	=
8	10
9	;
10	char
11	b
12	=
13	"abc"
14	;
15	int
16	{
17	20
18	;
19	char
20	d
21	=
22	"xyz"
23	int
24	n
25	}

(ii) Identify the Lexical error:

$x = 12ab;$

Error = $12ab$

Invalid numeric Constant, Identifiers cannot start with a digit.

d) $(a+ab)^*$

Accepted String:

ϵ

a

a b

a a

a a b.

e) $(a b a)^* + (a^* b^*)^*$

• ϵ

• a b a

• a b

• a a b b.

• a a a b b b.

4) Find a R.E for $\{0,1\}^*$:

(i) Even Numbers:

$$R.E = 1^* (01^* 01^*)^*$$

(ii) String Containing at least one.

$$R.E = (0+1)^* 1 (0+1)^*$$

(iii) String with both number of 0's & 1's are odd.

$$R.E = (00+11)^* (01+10) (00+11)^* (01+10) (00+11)^* (01+10) (00+11)^*)^*$$

(iv) String with 1 & end with 0.

$$R.E = 1 (0+1)^* 0.$$

(v) String that do not contain consecutive 1's.

$$R.E = (0+10)^* (\epsilon+1).$$

c) FSM state :

Tokens	Transition
int	3
Var123	6
=	1
123.45	5
;	1
float	5
Var456	6
=	1
123.45	6
;	1
if	2
(	1
Var123	2
=	6
Var456	1
)	1
{	6
return	1
;	6
}	1

$$= 3 + 6 + 1 + 5 + 1 + 5 + 6 + 1 + 6 + 1 + 2 + 1 + 6 + 2 + 6 + 1 + 1 + 6 + 1 + 1$$

$$= 70$$

Total FSM = 70.

5) a)

S.No	Token
1	int
2	Var 123
3	=
4	123 45
5	;
6	float
7	Var 456
8	=
9	123.45
10	;
11	if
12	(
13	Var 123
14	=
15	)
16	}
17	{
18	return
19	;
20	}

b) Categorized tokens :

1) Keyword :

int
float
if
return ,

2) Operators

+ =
* =
* = =

3) Separator

* ;
* ;
* (
*)
* {
* }
* ;

4) Identifiers

* Var 123
* Var 456
* Var 123
* Var 456.

5) Integers Literated

* 12345